

Propylene Glycol Safety and Environmental Considerations: A Buyer's Guide – RawSource

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Source Content

By RawSource Sourcing Desk, Commercial & Sourcing Desk · About RawSource A buyer asks for “the safe antifreeze,” signs off on a tote of propylene glycol, and assumes the safety question is closed. Then the QC lead asks which grade it is, the EHS manager asks why the SDS lists it as combustible, and the wastewater permit holder asks what its discharge does to the plant’s dissolved-oxygen numbers. Propylene glycol is one of the lower-hazard industrial fluids in routine use, but “low hazard” is not “no hazard,” and the questions that decide compliance sit in the details: grade, flash point, exposure route, and oxygen demand. This guide walks the safety and environmental profile of propylene glycol the way a sourcing team has to read it, with the numbers, the named regulations, and the trade-offs you can put in front of EHS and procurement at the same time. Is propylene glycol safe to handle? Within normal industrial controls, propylene glycol is one of the lower-hazard glycols a plant will handle. The FDA classifies it as Generally Recognized As Safe (GRAS) for direct food use under 21 CFR 184.1666. The body metabolizes propylene glycol to lactic and pyruvic acid, not to the toxic acids that make ethylene glycol dangerous. PubChem records low acute oral toxicity and a near-odorless, colorless liquid. Volatility is low. PubChem lists a vapor pressure of 0.08 mmHg at 68°F (20°C), so vapor inhalation at ambient temperature is a minor route. The exposure that matters is aerosol. Theatrical fog fluid, HVAC humidification, and deicing spray all put PG into the air as a fine mist, and inhaled mist can irritate the airways and eyes. That is the trade-off behind the “non-toxic” label: the molecule is benign to swallow in food doses but still an airway irritant when you aerosolize drums of it. Dermal contact is the second watchpoint. Propylene glycol is a recognized contact sensitizer in patch testing, so a fraction of workers and end users develop allergic dermatitis on repeated skin exposure. For most handling, nitrile gloves and splash goggles cover it. For misting or spraying operations, add local exhaust ventilation and treat the aerosol, not the liquid, as the hazard. Buyers formulating skin-contact products should weigh this against PG’s value as a humectant. The beauty and personal care sector uses it heavily for exactly that reason and manages the sensitization risk through concentration limits. What do the exposure and classification rules require? There is no OSHA permissible exposure limit (PEL) assigned to propylene glycol, and it carries no harmonized hazard classification under the EU’s CLP regulation through ECHA. Pure PG is also not regulated as a hazardous material for transport, so it ships without a UN number or hazmat placard. That light regulatory footprint is real, and it is one reason PG displaces harsher solvents in closed-loop systems. The absence of a PEL does not remove the paperwork. A Safety Data Sheet (SDS) is

still required, and for any grade-sensitive use you should demand a Certificate of Analysis (CoA) per lot. The CoA is where the safety story gets verified: assay, water content, and residual-glycol limits. Treat “no PEL, no UN number” as a reason to focus your diligence on grade and documentation, not on transport controls. Practical recommendation: write the grade, the assay floor, and the required CoA fields into the purchase specification before the first quote, not after the first shipment. The same molecule sold as a technical fluid and as a pharmaceutical excipient carries different documentation, and the gap is invisible until an auditor asks for it. How does propylene glycol compare with ethylene glycol on safety? The gap is large. Ethylene glycol is acutely toxic to humans and animals, while propylene glycol is not, and that single fact is why PG commands a price premium in food and pharmaceutical applications and in pet-contact antifreeze. Ethylene glycol metabolizes to glycolic and oxalic acid, which damage the kidneys, and a lethal dose for an adult is measured in tens of milliliters. Propylene glycol metabolizes to lactic and pyruvic acid, normal products of human energy metabolism.

Property	Propylene glycol (CAS 57-55-6)	Ethylene glycol (CAS 107-21-1)
Acute human toxicity	Low	High
Primary metabolite	Lactic / pyruvic acid	Glycolic / oxalic acid
FDA food status	GRAS (21 CFR 184.1666)	Not approved for food
Use near food / potable / pets	Suitable	Avoid
Relative freeze-protection efficiency	Slightly lower per unit	Slightly higher per unit
Relative cost	Higher	Lower

The decision is rarely about chemistry alone. Ethylene glycol is cheaper and a marginally more efficient freeze-point depressant, so it stays in sealed automotive and industrial loops where human contact is controlled. Propylene glycol wins the moment a leak could reach a food stream, a drinking-water line, or an animal. For the full breakdown of where each one belongs, see propylene glycol vs. ethylene glycol. The toxicity contrast is the entire commercial reason PG exists as a separate product line instead of a niche of the EG market. What are the environmental considerations for propylene glycol? The environmental profile is favorable on persistence and unfavorable on oxygen demand. Propylene glycol is readily biodegradable, does not bioaccumulate, and shows low direct toxicity to aquatic organisms. The problem is what biodegradation costs the water. As microbes break PG down, they consume dissolved oxygen, so a concentrated release exerts high biochemical oxygen demand (BOD) and chemical oxygen demand (COD). Drop enough PG into a slow stream and the oxygen crash, not the molecule, is what kills the fish. This is not a theoretical risk. Airport deicing is the textbook case: glycol-based fluids run off runways in winter, reach surface water, and depress dissolved oxygen downstream. The EPA addressed it directly with the Airport Deicing Effluent Guidelines, 40 CFR Part 449, finalized in 2012, which set discharge requirements for deicing operations. Any buyer moving PG at deicing or heat-transfer volumes should expect their wastewater permit to track BOD load. At 1.04 g/mL at 68°F (20°C), propylene glycol is slightly denser than water, but it is fully miscible, so a spill disperses into the water column instead of pooling on the surface or sinking as a recoverable layer. That changes spill response: you cannot skim it. Its high water solubility also makes it mobile in soil, with low sorption, so a ground spill can reach groundwater quickly. Contain PG before it reaches a storm drain, and keep concentrated product out of any discharge that feeds an oxygen-sensitive water body. For plants running PG as a heat-transfer or process fluid, the industrial manufacturing discharge question is the one to model early, because retrofitting BOD treatment after a permit exceedance is expensive. How does grade change the safety and compliance picture? Grade is the line between a legal food ingredient and an industrial fluid that must never touch a food stream, even though the CAS number on both drums reads 57-55-6. The molecule is identical; the impurity ceiling and the documentation are not.

Attribute	Industrial / technical	USP / EP (pharma)	Food (FCC)
Typical assay	High, looser tolerance	Tight, pharmacopeia limit	Tight, FCC limit
Residual DEG / EG limits	Not specified	Tested and capped	Tested and capped
Documentation	TDS, basic CoA	Full CoA, compendial testing	Full CoA, food-safety records
Permitted use	Coolants, solvents, antifreeze	Drugs, oral and topical excipients	Food, flavor, supplement carrier

The residual-glycol line in that table is the one with a body count behind it. Diethylene glycol (DEG) and ethylene glycol contamination of pharmaceutical-grade glycols caused the 2022 and 2023 contaminated-syrup poisonings that killed children across several

countries and triggered medical-product alerts from the World Health Organization. The contaminant was cheaper industrial glycol substituted for, or contaminating, pharmaceutical-grade material. The lesson for buyers is blunt: for any ingestible, injectable, or topical use, require a USP or FCC CoA that explicitly reports tested DEG and EG limits, and do not accept industrial grade against a pharmaceutical specification because the price is better. Recommendation: match the grade to the worst-case end use of the fluid, not the average case. If a single tank ever feeds a food-contact or skin-contact line, buy to that standard for the whole tank. How should you store and transport propylene glycol? Storage is straightforward, because the controls focus on moisture and contamination, not fire. Propylene glycol is combustible, not flammable: its flash point near 210°F (99°C) sits above the 199°F (93°C) GHS cutoff for flammable liquids, and it will not flash at ambient temperature. Its freeze point near -76°F (-60°C) means it will not solidify in any normal storage environment, and it tends to supercool rather than crystallize. The active risk is water. Propylene glycol is hygroscopic, so an open or poorly sealed container pulls in atmospheric moisture and drifts the assay downward, which matters for any spec-controlled use. Store it sealed in HDPE or stainless steel, away from strong oxidizers, since PubChem notes that on decomposition it emits acrid smoke and irritating fumes. For assay-critical USP stock, a nitrogen blanket and strict first-in-first-out rotation protect the water content. Sealed and dry, PG holds its specification well over typical shelf-life windows. Boiling point is 370.8°F (188°C), so thermal handling is benign in ordinary use; the decomposition and irritant-fume concern only appears at fire temperatures. When choosing among glycol-family fluids for a given service, note that close relatives behave differently. Dipropylene glycol shifts volatility and solvency, and the ether propylene glycol methyl ether is a genuinely flammable solvent with a far lower flash point. Do not carry PG's combustible-not-flammable assumption across to the glycol ethers. For the broader picture of where these fluids earn their place, the industrial applications of propylene glycol cover the heat-transfer, humectant, and carrier roles that drive most volume. Putting it in the purchase spec Buyers comparing industrial against USP or FCC grade can review specifications on the propylene glycol product page and request a quote with the grade, assay floor, and CoA requirements stated up front. Naming the residual-glycol limits and the end use in the RFQ is the cheapest insurance against a grade mismatch reaching the wrong line. Four lines in a specification settle most of the safety and environmental questions in this guide: the grade against its compendial standard (technical, USP, or FCC), the assay floor and maximum water content, the required CoA fields including tested DEG and EG limits, and the discharge plan for any stream that could carry PG to surface water. Set those before the first quote, and the EHS, QC, and wastewater reviews close themselves instead of stalling the order. Frequently asked questions Is propylene glycol toxic? Not at the levels of normal handling or approved food and pharmaceutical use. The FDA lists it as GRAS under 21 CFR 184.1666, and the body metabolizes it to lactic and pyruvic acid. The watchpoints are dermal sensitization in susceptible people and high-dose intravenous pharmaceutical exposure, not industrial contact. Is propylene glycol flammable? No. Its flash point sits near 210°F (99°C), above the 199°F (93°C) GHS threshold for flammable liquids, so it is classified and stored as a combustible liquid. It will still burn if heated past its flash point near an ignition source. Is propylene glycol bad for the environment? It is readily biodegradable and does not bioaccumulate, but it exerts high biochemical oxygen demand. Concentrated discharges, such as airport deicing runoff, can deplete dissolved oxygen in receiving water and stress aquatic life. The EPA regulates airport deicing effluent under 40 CFR Part 449. What is the difference between industrial and USP propylene glycol? Same molecule, CAS 57-55-6. USP and food (FCC) grades carry tighter impurity limits, including tested limits on diethylene glycol and ethylene glycol residuals, plus the documentation needed for ingestible, topical, or food-contact use. Can propylene glycol replace ethylene glycol antifreeze? Yes, wherever toxicity matters: food plants, HVAC loops near potable systems, and areas with pets or wildlife. It accepts a small efficiency and cost penalty in exchange for much lower toxicity. Methodology: physical constants (boiling point, flash point, vapor pressure, density, freeze point) are drawn from PubChem CID 1030; food status from 21 CFR 184.1666; deicing discharge regulation from EPA 40 CFR Part 449. Grade

and contamination guidance reflect published FDA and WHO pharmaceutical-product alerts. Confirm lot-specific values against the supplier CoA before specifying. Frequently asked questions Is propylene glycol toxic? Is propylene glycol flammable? Is propylene glycol bad for the environment? What is the difference between industrial and USP propylene glycol? Can propylene glycol replace ethylene glycol antifreeze? Sources & methodology Regulatory citations are current as of publication. Chemical identities verified by CAS number against the RawSource catalog. Search 1,300+ industrial chemicals by name or CAS, or send us your spec — we quote by the drum, tote, or container. Browse the Chemical Index → Request a Quote